

Marquee — AI Design Canvas

A viewing-discovery agent for a streaming catalogue · Aima Alakhume · BrainStation AI Certification, Final Project (Option 3)

UNOFFICIAL STUDENT CONCEPT · NOT AFFILIATED WITH DISNEY

01

Agent persona & task scoping

Persona. A catalogue specialist, not a chatbot. It speaks plainly, it never bluffs, and it would rather say what it could not do than answer something adjacent.

In scope. Turning a plain-sentence mood, constraint or occasion into titles that exist in this catalogue, with a reason attached.

Out of scope. Anything that is not finding something to watch. A request with no viewing intent gets one sentence and a redirect, not a guess.

Goal. Cut the time between opening the app and pressing play — measured by decisions reached, not clicks produced.

Where it runs. A Chrome extension on the live service. It reads the rows already on the page and writes one new row above them, so the answer arrives where the decision is being made.

02

The users

A profile is **data, not a prompt**: watch history, weighted affinities over genres, tone tags and brands, and a set of negative weights. Switching profile changes the arithmetic, never the code path.

Same title, three profiles	Just me	Movie night	Date night
Tokyo Revengers anime, serialised, intense	14.0	4.4	2.2
Call It Love K-drama, subtitled, slow-burn	8.5	−4.4	6.3
Thor: Ragnarok superhero, comedic	2.6	9.8	3.2
Elemental romance, tender, animated	4.5	2.9	7.5

Kids mode is deliberately not a profile. It is a toggle any household can switch on, and a request can switch on for itself ("kid-friendly detective shows"). Safety that lives inside one persona is safety you lose the moment someone picks another.

03

System contract

Input	One sentence · the active profile · optionally a refinement of the last answer
Output	Exactly one of: ranked picks with grounded reasons · one clarifying question · a refusal naming the rule that refused
Invariants	Every title returned exists in the catalogue by id · at most two clarifying turns · no constraint is ever loosened silently
Stop	Picks delivered · question asked · refusal issued · catalogue exhausted after the published relaxation order
Model	Optional and off by default. Two hooks only — fill gaps in the parse, re-order a shortlist. Both fall back to the deterministic core on any failure, timeout or empty answer

05

Knowledge stack & tools

- Catalogue** — 315 titles that stream on Disney+ or Hulu in the US: 148 films, 151 series, 16 shorts. Runtime, certificate, genre and artwork come from TMDB; availability comes from JustWatch through TMDB.
- Two tiers, recorded not hidden** — 115 titles have a tone layer written and checked by hand; 200 have one derived by rule. Each row carries its `tier`, and hand-checked titles score slightly higher.
- Tone layer** — `energy` and `warmth` (1–5) plus tone tags. This is the part no metadata source supplies, and it is what makes "comforting" a query rather than a wish.
- Profiles** — history, affinities, negative signals, hard filters. Synthetic, and labelled so.
- Tools** — `retrieve()` structured filter · `rank()` weighted scoring · `validate()` catalogue membership.
- Model** — **optional, on-device, two narrow jobs** — Llama 3.2 3B runs in the browser on WebGPU, downloaded once and never sent anything. It fills gaps in the parse and re-orders a shortlist. It never selects from the full catalogue and never has the last word.
- Three tiers of answer** — on-device model → cached build-time parse → deterministic core. The panel always states which one answered.

04

Logic & flow

1 PARSE Sentence → structured request; confidence scored	2 CLARIFY One question, or proceed and say so. Cap: 2 turns	
3 RETRIEVE Deterministic filter; hard filters first	4 RANK Weighted score, then optional model re-rank on tone	5 VALIDATE Membership check; invented ids dropped
6 EXPLAIN Reason from metadata + this profile's history	7 REFINE Re-enters at step 3 with locked constraints	
EXIT · REFUSES A household rule removes the whole request. It names the rule and offers the nearest thing it is allowed to offer.		
EXIT · RELAXES Nothing matches. Constraints are surrendered in a published order — runtime, year, format, then genre, with the subject asked for surrendered last — and the viewer is told which one went.		
EXIT · CATCHES A title the model invented is dropped before display; the step re-runs on what survived.		

06

Guardrails & failure handling

- Safety is structural, not instructional.** The audience filter runs inside retrieval, before ranking. "Ignore the parental settings" is not actionable because no unsuitable title was ever a candidate.
- A rating is not an audience.** `kid_safe` is modelled separately from `certification`. Age bands carry a certificate allow-list *and* an intensity ceiling, because a TV-Y7 action series is rated for seven-year-olds and is still wrong for a five-year-old.
- The model cannot say who is watching.** `company` and `age_band` are not in its schema and are stripped if returned. It can add to a request; it can never widen one.
- Everything the model returns is re-validated.** Genres, tags and numbers are checked against the catalogue's own vocabulary; re-ranked ids are checked for membership. Invented values are dropped before display.
- Honest degradation.** Every relaxation is named, in a published order: `runtime` → `year` → `format` → `genre_mode` → `genres` → `text`. Subject terms are surrendered last.
- Honest vocabulary.** A word the lexicon cannot read is reported; a word the model resolved is credited to the model rather than reported as unused.
- Locked refinements · turn cap · scope guard.** A constraint just set cannot be undone by the fallback path; no clarification loop beyond two turns; no viewing intent gets one line and a redirect.
- No-model path is the default.** Every guardrail runs without a model, so the agent degrades to deterministic rather than to nothing.

07

Evaluation & optimisation

42 cases: happy path, ambiguity, empty results, child safety, prompt injection, out-of-scope, contradiction, refinement, negation, genre enforcement, format, the quiz path, vocabulary. Constraint satisfaction and grounding are checked mechanically on every case; taste fit is scored 1–5 by hand. A further 6 cases test the model seam against a stub, so the safety claims are proven rather than asserted.

Run	Passing	What the run taught
v1	9 / 15	Two lexicon gaps; three design faults — a filter that only scored, a fallback that overrode an explicit instruction, a cache that outranked the user
v12	39 / 39	Growing the catalogue from 115 to 315 falsified two passing tests. A test that passes because the data is small is not a test
v15	42 / 42	An episode length read as a mood; tension and gentleness both true at once; a certificate that could never have caught <i>Gargoyles</i> ; a re-rank prompt a small model misread

Full log, all fifteen rounds: `docs/iteration_log.md`. Harness: `build/run_evals.mjs`, `build/test_model_seam.mjs`.